

MedVirtuoso – Complete System Architecture & Workflow Report

The platform is designed to streamline medical imaging uploads, viewing, processing, reporting, and delivery using centralized cloud infrastructure.

Objectives of the Platform

- Provide a centralized web-based medical imaging workflow system.
- Allow clients to upload patient imaging studies and track progress.
- Enable operators to process and manage uploaded studies.
- Integrate secure cloud storage for DICOM and imaging files.
- Provide secure report generation and download capabilities.
- Maintain workflow transparency through live status updates.

User Roles

- Client/User – Doctors, hospitals, or institutions uploading studies.
- Operator – MedMarvel staff handling processing and report uploads.

Authentication & Login System

- Role-based login system.
- Secure authentication and session management.
- Separate dashboards for clients and operators.

Client Dashboard – Worklist System

- Displays uploaded studies in a structured worklist.
- Contains patient details, study information, modality, status, and notes.
- Includes View, Edit, Delete, Upload, and Download Report actions.

Upload Workflow

- Manual patient/study entry.
- Direct DICOM/MRI/PET upload.
- Industry-standard imaging link upload option.

Operator Dashboard

- Access uploaded files or imaging links.
- Download studies for reporting team.
- Upload final reports.
- Manage study workflow statuses.

8. End-to-End Workflow

1. Client logs into MedVirtuoso.
2. Client uploads imaging files or imaging links.
3. Files are uploaded securely to AWS S3 cloud storage.
4. Operator accesses study from dashboard.

5. Status changes to 'Processing'.
6. Operator forwards study to reporting team.
7. Reporting team generates report.
8. Operator uploads completed report.
9. Status changes to 'Done'.
10. Client downloads report.

9. Technology Stack

Frontend:

- Next.js / React.js
- Tailwind CSS
- TypeScript

Backend:

- Node.js
- Express.js / Next.js APIs
- REST APIs

Database:

- PostgreSQL
- Prisma ORM

Cloud:

- AWS S3
- Docker
- Cloud Deployment

Authentication:

- JWT Authentication
- Encrypted Credentials

Medical Imaging:

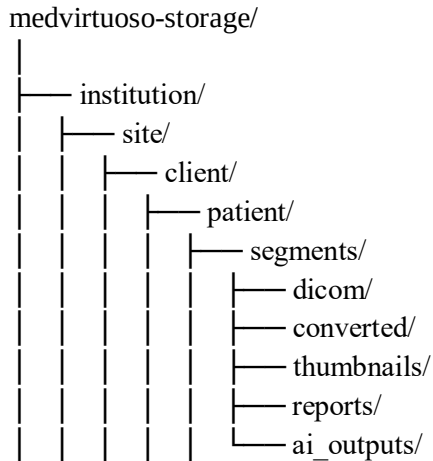
- MRICroGL Viewer Integration

10. AWS S3 Cloud Infrastructure & Data Hierarchy

MedVirtuoso uses AWS S3 cloud storage for secure storage and management of large DICOM and medical imaging datasets.

10.1 S3 Hierarchical Structure

Institution → Site → Client → Patient → Segments



10.2 Explanation of Hierarchy

Institution: Healthcare organization such as Apollo, Fortis, Medanta, etc.

Site: Specific hospital branch or diagnostic center.

Client: Doctor, staff..

Patient: Patient-specific folder.

Segments: Individual imaging studies or processed datasets.

10.3 File Upload Workflow

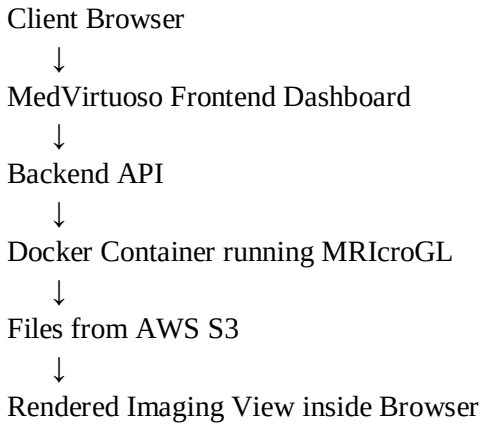
11. Frontend uploads imaging data to backend.
12. Backend validates files and metadata.
13. Files are uploaded securely to AWS S3.
14. Database stores S3 object paths and patient metadata.
15. Studies become visible on operator dashboard.

11. MRICroGL Viewer Containerization & Web Integration

MedVirtuoso integrates MRICroGL directly into the browser using Docker-based containerization technology.

- MRICroGL is traditionally a desktop application.
- Docker containers allow MRICroGL to run in isolated environments.
- Users can access imaging studies directly from the browser.
- No local software installation is required.

11.2 Viewer Architecture



11.3 Viewer Workflow

16. Client clicks View on dashboard.
17. Backend authenticates user.
18. Backend retrieves study paths from PostgreSQL.
19. Secure S3 URLs are generated.
20. Docker container loads imaging data.
21. MRICroGL renders imaging study.
22. Rendered view appears inside browser.

12. Status Management System

Status	Meaning
Uploaded	Files uploaded by client.
Processing	Operator has started processing.
Done	Final report uploaded and ready.

Client Browser → Frontend Dashboard → Backend API → Database + AWS S3 Storage → Operator Dashboard → Reporting Team → Report Upload → Client Download

The MedVirtuoso platform architecture has been defined across frontend, backend, cloud infrastructure, medical imaging workflows, operator coordination, and viewer integration.